

MECHANISMS AND BEHAVIORS IN COMPLEX SYSTEMS WITH GROUP INTERACTIONS

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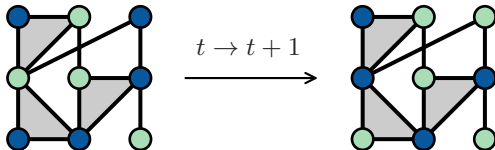
February 28, 2025

SOME REFERENCES

- Battiston, F. *et al.* The physics of higher-order interactions in complex systems. *Nature Physics* **17**, 1093–1098 (2021)
- Rosas, F. E. *et al.* Disentangling high-order mechanisms and high-order behaviours in complex systems. *Nature Physics* **18**, 476–477 (2022)
- **Robiglio, T. *et al.* Synergistic signatures of group mechanisms in higher-order systems. *arXiv preprint arXiv:2401.11588* (2024)**
- Peel, L. *et al.* Statistical inference links data and theory in network science. *Nature Communications* **13**, 6794 (2022)

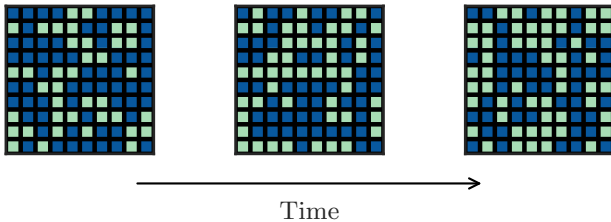
(a)

Mechanism



(b)

Behavior



MECHANISMS

Structural and dynamical rules controlling the causal evolution of the system.

Examples:

$$H = - \sum_{i \neq j} J_{ij} s_i s_j, \quad P(s_i^t \rightarrow s_i^{t+1}) = \dots$$

$$\rho_i^{t+1} = \rho_i^t + \mu[1 - \rho_i^t] - \rho_i^t \left\{ 1 - \prod_j \left[1 - \beta A_{ij}(1 - \rho_j^t) \right] \right\}$$

$$\dot{\theta}_i = \omega_i + \lambda \sum_j A_{ij} \sin(\theta_j - \theta_i)$$

BEHAVIORS

Measurable observables quantifying statistical interdependencies between units.

Examples:

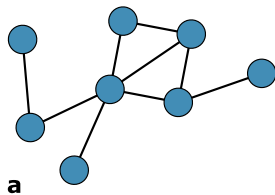
- Correlations
- Mutual information
- Autocorrelation
- Transfer entropy
- ...

Beyond pairwise?

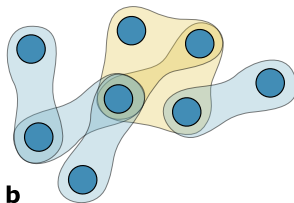
HIGHER-ORDER MECHANISMS

Explicit representations of group interactions^{1,2}:

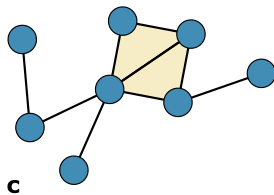
Network



Hypergraph



Simplicial complex

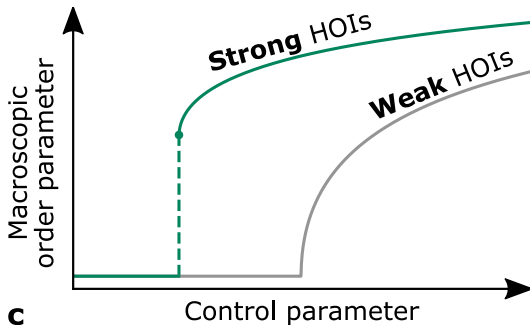


¹Battiston, F. *et al.* Networks beyond pairwise interactions: Structure and dynamics. *Physics reports* **874**, 1–92 (2020).

²Battiston, F. *et al.* The physics of higher-order interactions in complex systems. *Nature Physics* **17**, 1093–1098 (2021).

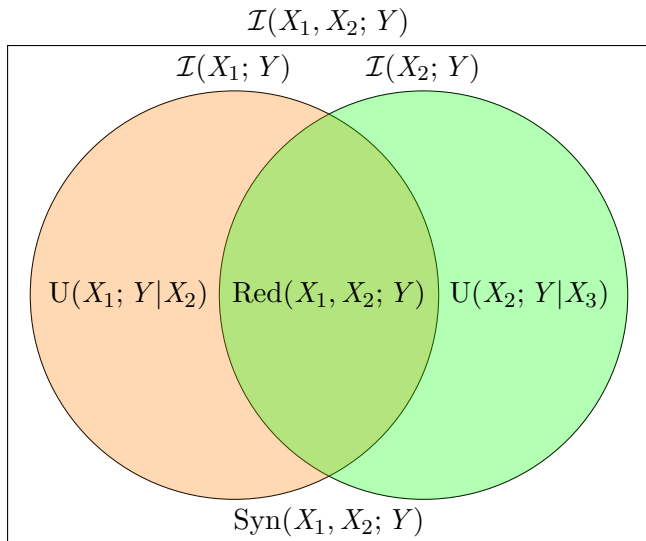
HIGHER-ORDER MECHANISMS

Higher-order interactions lead to explosive phenomena:



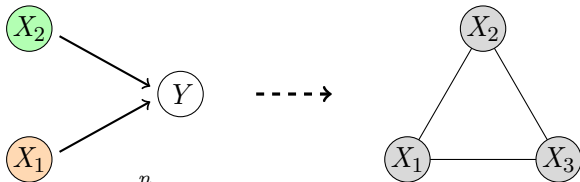
HIGHER-ORDER BEHAVIORS

Information two r.v. X_1 and X_2 contain a third r.v. Y ³:



³Williams, P. L. & Beer, R. D. Nonnegative decomposition of multivariate information. *arXiv preprint arXiv:1004.2515* (2010).

HIGHER-ORDER BEHAVIORS



$$\text{TC}(\mathbf{X}^n) \equiv \sum_{j=1}^n \mathcal{H}(X_j) - \mathcal{H}(\mathbf{X}^n) \rightarrow \text{Collective constraints}$$

$$\text{DTC}(\mathbf{X}^n) \equiv \mathcal{H}(\mathbf{X}^n) - \sum_{j=1}^n \mathcal{H}(X_j | \mathbf{X}_{-j}^n) \rightarrow \text{Shared randomness}$$

$$\text{O-information: } \Omega(\mathbf{X}^n) \equiv \text{TC}(\mathbf{X}^n) - \text{DTC}(\mathbf{X}^n)$$

Intuitively⁴:

$\Omega(\mathbf{X}^n) > 0 \rightarrow \text{Redundancy dominates}$

$\Omega(\mathbf{X}^n) < 0 \rightarrow \text{Synergy dominates}$

⁴Rosas, F. E. *et al.* Quantifying high-order interdependencies via multivariate extensions of the mutual information. *Physical Review E* **100**, 032305 (2019).

LOW-ORDER MECHANISM

LOW-ORDER BEHAVIOR

Correlations, MI, Granger caus., *etc.*

HIGHER-ORDER MECHANISM

HIGHER-ORDER BEHAVIOR

TC, O-info, *etc.*

**LOW-ORDER
MECHANISM**



**LOW-ORDER
BEHAVIOR**

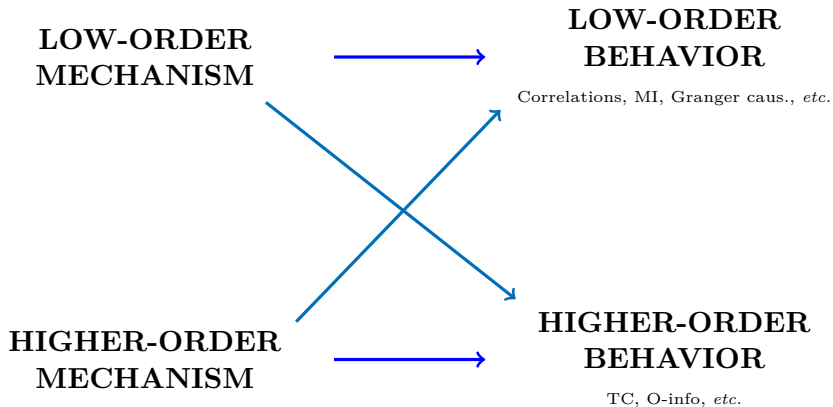
Correlations, MI, Granger caus., *etc.*

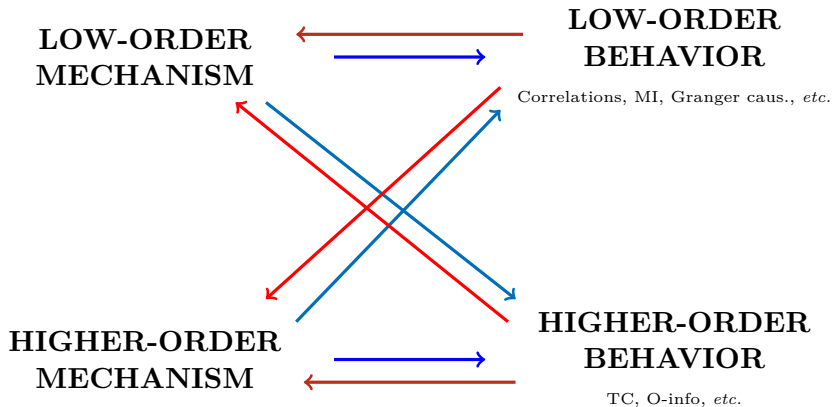
**HIGHER-ORDER
MECHANISM**



**HIGHER-ORDER
BEHAVIOR**

TC, O-info, *etc.*

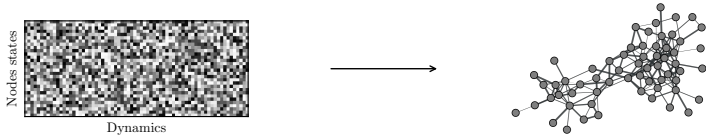




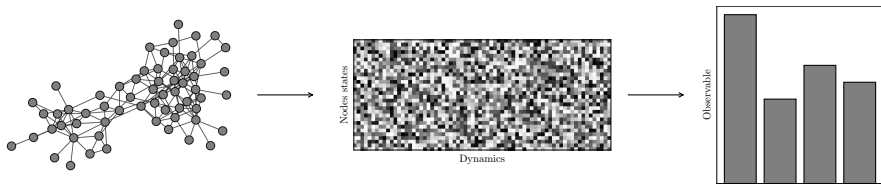
WHAT TO DO?

1. **Clarify definitions**
2. **Systematic exploration**
3. Full mapping?

THE HARD TASK

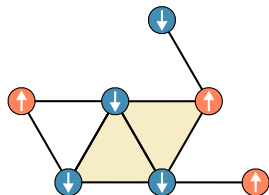


THE “EASY” TASK



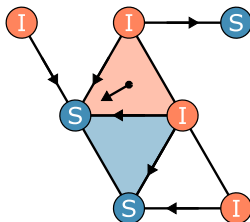
THE “EASY” TASK - MODELS

(i) Simplicial Ising model⁵



$$H = -J_0 \sum_{i=1}^N S^i - \sum_{\ell=1}^{\ell_{\max}} J_{\ell} \sum_{\{\sigma \in \mathcal{K}: |\sigma|=\ell\}} \left[2 \bigotimes_{i \in \sigma} S^i - 1 \right]$$

(ii) Simplicial contagion model⁶



Pairwise infection rate: β_1

Group infection rate: β_2

Recovery rate: μ

⁵Robiglio, T. *et al.* Higher-order Ising model on hypergraphs. *arXiv preprint arXiv:2411.19618* (2024).

⁶Iacopini, I. *et al.* Simplicial models of social contagion. *Nature communications* **10**, 2485 (2019).

THE “EASY” TASK - BEHAVIOR OBSERVABLE

Generalize O-information to time-lagged correlations⁷:

1. Considering n variables $\mathbf{X} = (X_1, \dots, X_n)$ on which we compute the O-information Ω_n ,
2. Add a new variable Y ,
3. Compute the variation of O-information:

$$\Delta_n = \Omega_{n+1}(\mathbf{X}, Y) - \Omega_n(\mathbf{X})$$

4. Condition of the past of Y :

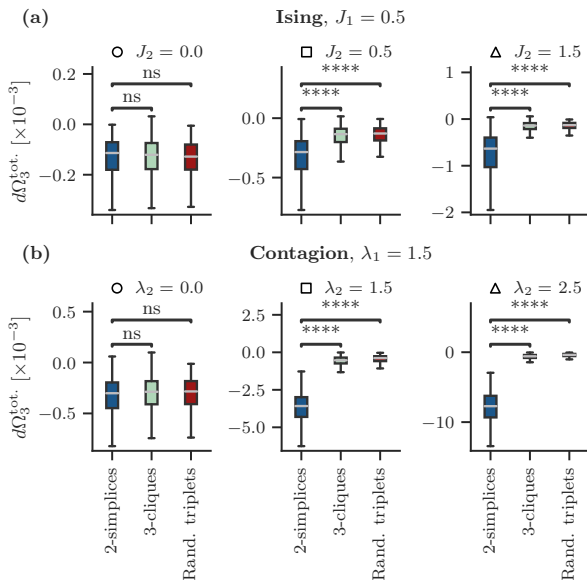
$$d\Omega_n(Y; \mathbf{X}) \equiv (1 - n) \mathcal{I}(Y; \mathbf{X} | Y_0) + \sum_{j=1}^n \mathcal{I}(Y; \mathbf{X}_{-j} | Y_0)$$

We define **total** dynamical O-information:

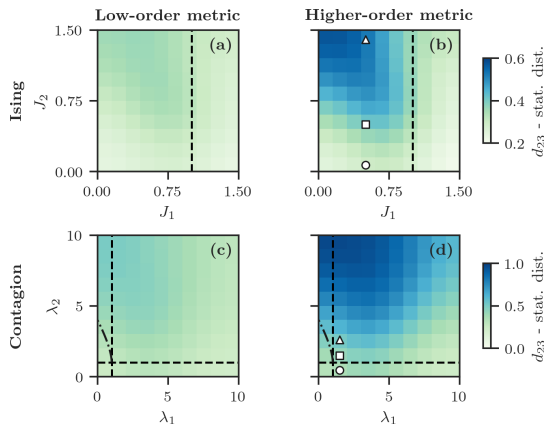
$$d\Omega_n^{\text{tot.}}(\mathbf{X}) \equiv \sum_{j=1}^n d\Omega_{n-1}(X_j; \mathbf{X}_{-j})$$

⁷Stramaglia, S. *et al.* Quantifying dynamical high-order interdependencies from the o-information: an application to neural spiking dynamics. *Frontiers in Physiology* 11, 595736 (2021).

SYNERGISTIC SIGNATURES



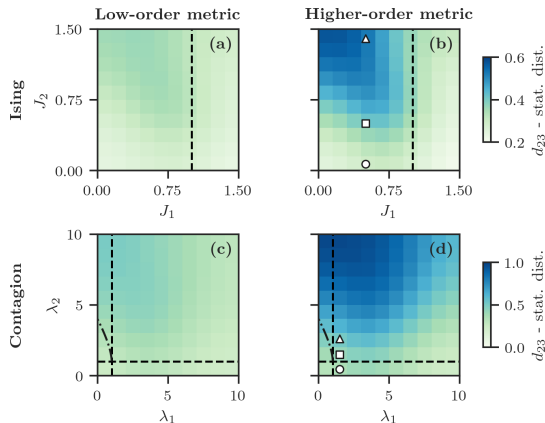
INSUFFICIENCY OF LOW-ORDER METRICS



Transfer entropy: $\mathcal{T}(X; Y) \equiv \mathcal{I}(X_t, Y_{t+1} | Y_t)$

Statistical distance: $d(P_2, P_3) = \frac{1}{2} \sum_{x \in \mathcal{X}} |P_2(x) - P_3(x)|$

LANDSCAPE



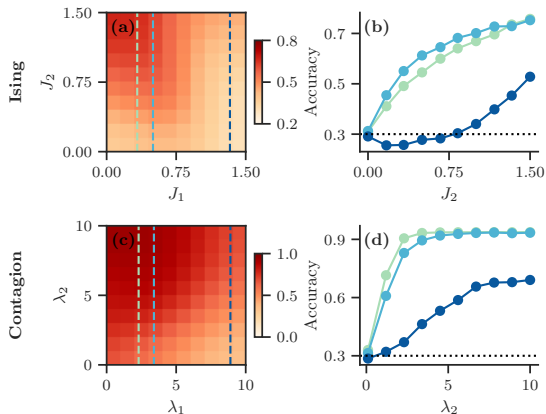
Pairwise Ising: $J_1^{\text{cr.}} = T/\langle k_1 \rangle$

Contagion: pairwise $\lambda_1^{\text{cr.}} = 1$, disc. transition $\lambda_1 = 2\sqrt{\lambda_2} - \lambda_2$

DOWNSTREAM TASKS

Given: pairwise structure⁸ and $\langle k_2 \rangle$

Task: find the true higher-order interactions



⁸Young, J.-G. *et al.* Hypergraph reconstruction from network data. *Communications Physics* 4, 135 (2021).

SUMMARY

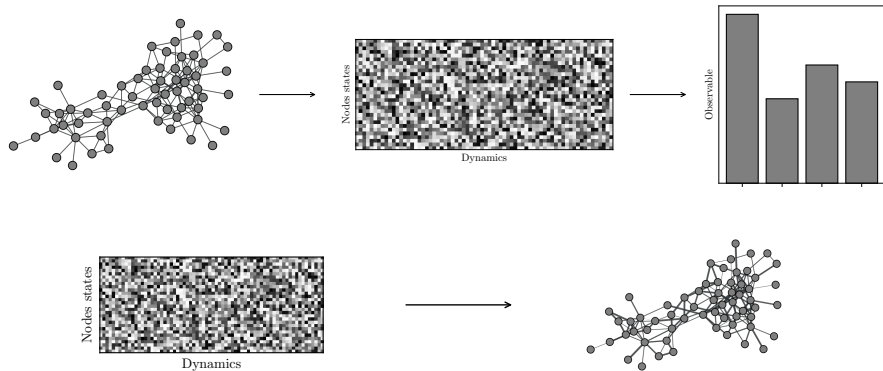
- Don't mix behaviors and mechanisms
- Emergence of synergistic signatures of higher-order mechanisms
- Higher-order signatures are invisible to low-order observables
- These signatures are a “weak” emergent phenomena
- Can be useful for downstream tasks (e.g. detection)

General picture:

Intricate relation between mechanisms and behaviors.

What can be done better?

ADDRESS THE HARD TASK

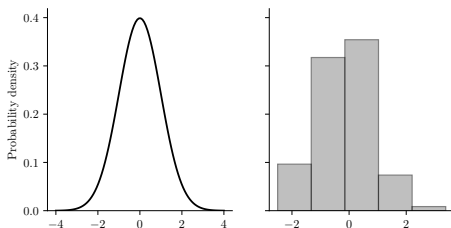


HOW DO WE ACTUALLY COMPUTE THE OBSERVABLES?

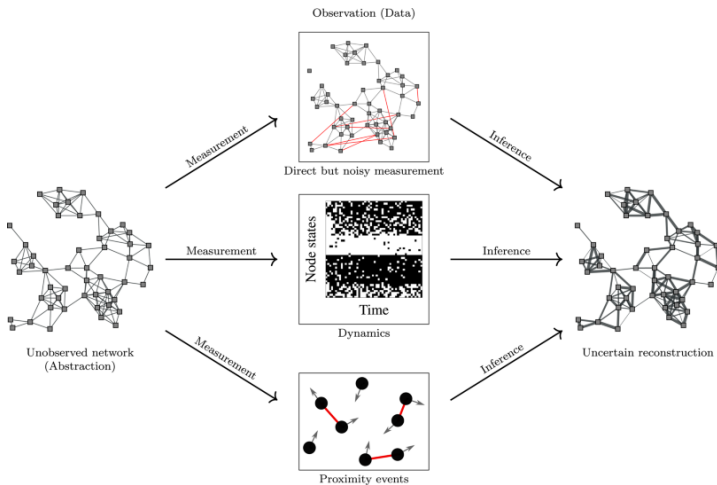
$$\mathcal{H}(X) = - \sum_{x \in \Sigma} p(x) \log p(x)$$

Given a set of M observations $\{x_i\} \in \{0, 1\}^M$:

$$\mathcal{H}(X) = - \sum_{x \in \{0,1\}} p_x \log p_x \text{ where } p_x = \frac{1}{M} \sum_{i=1}^M \delta_{x_i, x}$$



INVERSE PROBLEM⁹



⁹Peel, L. *et al.* Statistical inference links data and theory in network science. *Nature Communications* 13, 6794 (2022).

BAYESIAN APPROACH

We can only make hypotheses and evaluate to which extent the data supports them.

$$P(\mathbf{A}|\mathbf{D}) = \frac{P(\mathbf{D}|\mathbf{A})P(\mathbf{A})}{P(\mathbf{D})}$$

or if we have a model for the dynamics¹⁰ with parameters $\{d\}$:

$$P(A, \{d\}|\mathbf{D}) = \frac{P(\mathbf{D}|\mathbf{A}, \{d\})P(\mathbf{A}, \{d\})}{P(\mathbf{D})}$$

¹⁰Peixoto, T. P. Network reconstruction and community detection from dynamics. *Physical review letters* **123**, 128301 (2019).

WRAPPING UP

- “Network” language can be *too intuitive* → fix terminology behaviors/mechanisms
- Relation between synergistic behavior and group mechanisms
- No “model-free” analysis

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